



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2004
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK # 2

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Write using black or blue pen.
- Board approved calculators may be used.
- All *necessary* working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Hand in your answer booklets in **3** sections.
Section A (Questions 1 - 3), Section B (Questions 4 - 5) and Section C (Questions 6 - 7).
- Start each **NEW** section in a separate answer booklet.

Total Marks - 75 Marks

- Attempt Sections A - C
- All questions are NOT of equal value.

Examiner: *E. Choy*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Total marks – 78

Attempt Questions 1 – 7

All questions are NOT of equal value

Answer each section in a SEPARATE writing booklet. Extra writing booklets are available.

SECTION A (Use a SEPARATE writing booklet)

Question 1 (15 marks)		Marks
(a)	Evaluate $\int_0^3 \frac{x dx}{\sqrt{16+x^2}}$	3
(b)	By completing the square first, find $\int \frac{dx}{x^2+6x+13}$	2
(c)	Use integration by parts to find $\int x e^{-x} dx$	2
(d)	Find $\int \cos^3 \theta d\theta$	3
(e)	(i) Find real numbers A , B , and C such that $\frac{x^2-4x-1}{(1+2x)(1+x^2)} = \frac{A}{1+2x} + \frac{Bx+C}{1+x^2}$	3
	(ii) Hence find $\int \frac{x^2-4x-1}{(1+2x)(1+x^2)} dx$	2

Question 2 (10 marks)

Marks

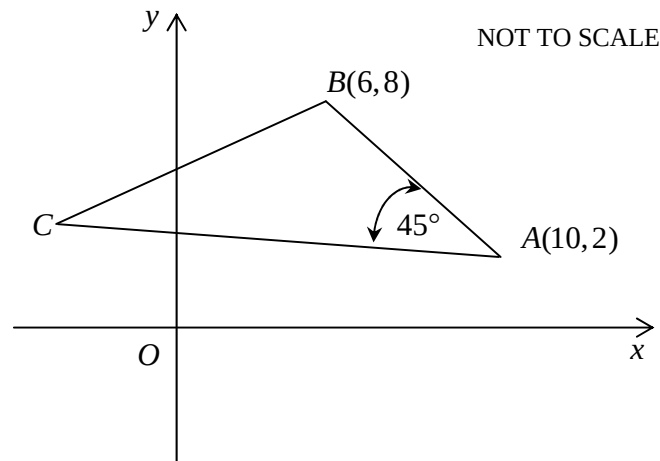
(a) On separate Argand diagrams, sketch the locus defined by:

(i) $2|z| = z + \bar{z} + 2$ 2

(ii) $|z^2 - (\bar{z})^2| \geq 4$ 2

(iii) $\arg(z-1) - \arg(z+1) = -\pi/3$ 2

(b)



$\triangle ABC$ is drawn in the Argand diagram above where $\angle BAC = 45^\circ$, A and B are the points $(10, 2)$ and $(6, 8)$ respectively.
The length of side AC is twice the length of side AB .

Find:

(i) the complex number that the vector $\overrightarrow{AB}$ represents the complex number $-4 + 6i$; 2

(ii) the complex number that the point C represents. 2

Question 3 (12 marks)

(a) The quadratic equation $x^2 - x + K = 0$, where K is a real number, has two distinct positive real roots α and β .

(i) Show that $0 < K < \frac{1}{4}$ 1

(ii) Show that $\alpha^2 + \beta^2 = 1 - 2K$ and deduce that $\alpha^2 + \beta^2 > \frac{1}{2}$ 2

(iii) Show that $\frac{1}{\alpha^2} + \frac{1}{\beta^2} > 8$ 2

SECTION A continued

Question 3 continued

Marks

- | | | | |
|-----|-------|---|---|
| (b) | (i) | Show, using De Moivre's Theorem, that $z = \omega$, where $\omega = \sqrt{2} + i\sqrt{2}$ satisfies $z^4 = -16$.
Hence write down, in the form $x + iy$ where x and y are real, all the other solutions of $z^4 = -16$. | 3 |
| | (ii) | Hence write $z^4 + 16$ as a product of two quadratic factors with real coefficients. | 2 |
| | (iii) | Show that $\omega + \frac{\omega^3}{4} + \frac{\omega^5}{16} + \frac{\omega^7}{64} = 0$ | 2 |

SECTION B (Use a SEPARATE writing booklet)

Question 4 (8 marks)

Marks

(a) Evaluate

(i) $\int_0^a x\sqrt{a-x} \, dx$ 2

(ii) $\int_0^1 \frac{\sin^{-1} x}{\sqrt{1+x}} \, dx$ 2

(b) (i) Using the substitution $t = \tan \frac{x}{2}$ show that 2

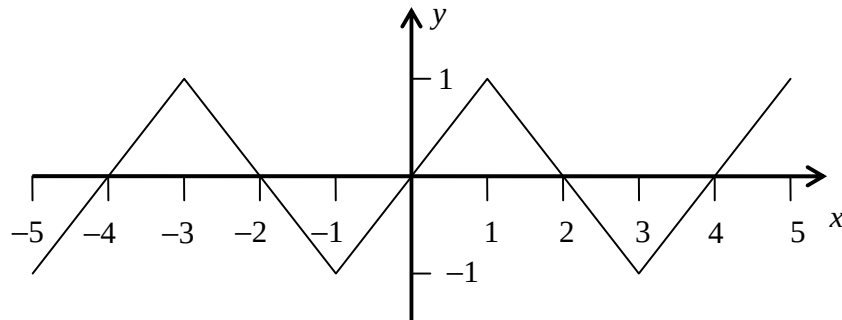
$$\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \cos x + \sin x} = \ln 2$$

(ii) Hence, by substituting $u = \frac{\pi}{2} - x$ evaluate 2

$$\int_0^{\frac{\pi}{2}} \frac{x \, dx}{1 + \cos x + \sin x}$$

Question 5 (11 marks)

(a)



The diagram is a sketch of the function $y = h(x)$ for $-5 \leq x \leq 5$.

On separate diagrams sketch each of the following:

(i) $y = h(x+1)$ 1

(ii) $y = \frac{1}{h(x)}$ 2

(iii) $y = h(|x|)$ 1

(iv) $y = \sqrt{h(x)}$ 2

(v) $y = h(\sqrt{x})$ 2

(b) Sketch the curve $9y^2 = x(x-3)^2$ showing clearly the coordinates of any turning point. 3

SECTION C (Use a SEPARATE writing booklet)

Question 6 (9 marks)

Marks

A firework missile of mass 0.2 kg is projected vertically upwards from rest by means of a force that decreases uniformly in 2 seconds from $2g$ newtons to zero and thereafter ceases. Assume no air resistance and that g is the acceleration due to gravity.

- (i) If the missile has an acceleration of $a \text{ m/s}^2$ at time t seconds, show that 3

$$a = \begin{cases} g(9-5t) & t \leq 2 \\ -g & t > 2 \end{cases}$$

[Hint: Draw a diagram showing the forces on the missile.]

- (ii) Hence find:
- (α) the maximum speed of the missile; 3
- (β) the maximum height reached by the missile. 3

Question 7 (10 marks)

Marks

A particle of mass 1 kg is projected from a point O with a velocity u m/s along a smooth horizontal table in a medium whose resistance is Rv^2 newtons when the particle has velocity v m/s. R is a constant, with $R > 0$.

- (i) Show that the equation of motion governing the particle is given by 1

$$\ddot{x} = -Rv^2$$

where x is the horizontal distance travelled from O .

- (ii) Hence show that the velocity, v m/s, after t seconds is given by 3

$$t = \frac{1}{R} \left(\frac{1}{v} - \frac{1}{u} \right)$$

An equal particle is projected from O simultaneously with the first particle, but vertically upwards with velocity u m/s in the SAME medium.

- (iii) Show that the equation of motion governing the second particle is given by 1

$$\ddot{y} = -(g + Rv^2)$$

where g m/s² is the acceleration due to gravity and y represents the vertical distance from O where the particle has a velocity of v m/s.

- (iv) Hence show that the velocity V m/s of the first particle when the second one is momentarily at rest is given by 5

$$\frac{1}{V} = \frac{1}{u} + \frac{1}{a} \tan^{-1} \left(\frac{u}{a} \right), \text{ where } Ra^2 = g$$

THIS IS THE END OF THE PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$



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Mathematics Extension 2

Sample Solutions

Section	Marker
A	Mr Hespe
B	Mr Parker
C	Mr Kourtesis

3 1. (a) Method 1:

$$\begin{aligned}
 I &= \int_0^3 \frac{xdx}{\sqrt{16+x^2}}, & \text{put } u &= 16+x^2 \\
 & & du &= 2xdx \\
 &= \frac{1}{2} \int_{16}^{25} u^{-\frac{1}{2}} du, & \text{when } x &= 0, & u &= 16 \\
 & & x &= 3, & u &= 25 \\
 &= \frac{1}{2} \left[2u^{\frac{1}{2}} \right]_{16}^{25}, \\
 &= 5 - 4, \\
 &= 1.
 \end{aligned}$$

Method 2:

$$\begin{aligned}
 I &= \int_4^5 \frac{udu}{u}, & \text{put } u^2 &= 16+x^2 \\
 & & 2udu &= 2xdx \\
 &= u \Big|_4^5, & \text{when } x &= 0, & u &= 4 \\
 & & x &= 3, & u &= 5 \\
 &= 1.
 \end{aligned}$$

Method 3:

$$\begin{aligned}
 I &= \int_0^{\tan^{-1} \frac{3}{4}} \frac{4 \tan \theta \cdot 4 \sec^2 \theta d\theta}{4 \sec \theta}, & \text{put } x &= 4 \tan \theta \\
 & & dx &= 4 \sec^2 \theta d\theta \\
 &= 4 \sec \theta \Big|_0^{\tan^{-1} \frac{3}{4}}, & \text{when } x &= 0, & \theta &= 0 \\
 &= 4 \left\{ \frac{5}{4} - 1 \right\}, & x &= 3, & \theta &= \tan^{-1} \frac{3}{4} \\
 &= 1.
 \end{aligned}$$

Method 4:

$$\begin{aligned}
 I &= \frac{1}{2} \int_0^9 \frac{du}{\sqrt{16+u}}, & \text{put } u &= x^2 \\
 & & du &= 2xdx \\
 &= \left[\frac{1}{2} \times 2 \times \sqrt{16+u} \right]_0^9, & \text{when } x &= 0, & u &= 0 \\
 &= 5 - 4, & x &= 3, & u &= 9 \\
 &= 1.
 \end{aligned}$$

Method 5:

$$\begin{aligned}
 I &= \frac{1}{2} \int_0^3 \frac{d(x^2)}{\sqrt{16+x^2}}, \\
 &= \left[\frac{1}{2} \times 2 \times \sqrt{16+x^2} \right]_0^3, \\
 &= 5 - 4, \\
 &= 1.
 \end{aligned}$$

$$\begin{aligned}
\boxed{2} \quad (b) \quad I &= \int \frac{dx}{(x^2 + 6x + 9) + 13 - 9}, \\
&= \int \frac{dx}{(x+3)^2 + 4}, \\
&= \frac{1}{2} \tan^{-1} \left(\frac{x+3}{2} \right) + C.
\end{aligned}$$

$$\begin{aligned}
\boxed{2} \quad (c) \quad I &= \int x e^{-x} dx, & \begin{array}{ll} u = x, & v' = e^{-x} \\ u' = 1, & v = -e^{-x} \end{array} \\
&= -x e^{-x} + \int e^{-x} dx, \\
&= -x e^{-x} - e^{-x} + C.
\end{aligned}$$

$$\begin{aligned}
\boxed{3} \quad (d) \quad \text{Method 1:} \\
I &= \int \cos^2 \theta \cdot \cos \theta d\theta, & \begin{array}{l} \text{put } \sin \theta = u \\ \cos \theta d\theta = du \end{array} \\
&= \int (1 - \sin^2 \theta) \cdot \cos \theta d\theta, \\
&= \int (1 - u^2) du, \\
&= u - \frac{1}{3} u^3 + C, \\
&= \sin \theta - \frac{1}{3} \sin^3 \theta + C.
\end{aligned}$$

Method 2:

$$\begin{aligned}
I &= \int \cos^2 \theta \cdot \cos \theta d\theta, \\
&= \int (1 - \sin^2 \theta) \cdot d \sin \theta, \\
&= \sin \theta - \frac{1}{3} \sin^3 \theta + C.
\end{aligned}$$

Method 3:

$$\begin{aligned}
I &= \int \cos^2 \theta \cdot \cos \theta d\theta, & \begin{array}{ll} u = \cos^2 \theta, & v' = \cos \theta \\ u' = -2 \sin \theta \cos \theta, & v = \sin \theta \end{array} \\
&= \cos^2 \theta \sin \theta + 2 \int \sin^2 \theta \cos \theta d\theta, \\
&= \cos^2 \theta \sin \theta + 2 \int \cos \theta (1 - \cos^2 \theta) d\theta, \\
&= \cos^2 \theta \sin \theta + 2 \sin \theta - 2 \int \cos^3 \theta d\theta, \\
3I &= \cos^2 \theta \sin \theta + 2 \sin \theta + c, \\
I &= \frac{1}{3} \{ \cos^2 \theta \sin \theta + 2 \sin \theta \} + C.
\end{aligned}$$

3

(e) (i) Method 1:

$$x^2 - 4x - 1 = A(1 + x^2) + (Bx + C)(1 + 2x).$$

$$\text{Put } x = -\frac{1}{2},$$

$$\frac{1}{4} + 2 - 1 = A(1\frac{1}{4}),$$

$$A = 1.$$

$$\text{Also, } x^2 - 4x - 1 = x^2(A + 2B) + x(B + 2C) + (A + C),$$

$$\text{so } A + 2B = 1,$$

$$1 + 2B = 1,$$

$$2B = 0,$$

$$B = 0.$$

$$\text{And } B + 2C = -4,$$

$$2C = -4,$$

$$C = -2.$$

Method 2:

$$\frac{x^2 - 4x - 1}{(1 + 2x)(1 + x^2)} = \frac{A}{1 + 2x} + \frac{Bx + C}{x^2 + 1}.$$

$$\lim_{x \rightarrow -\frac{1}{2}} \left\{ \frac{x^2 - 4x - 1}{1 + x^2} \right\} = \lim_{x \rightarrow -\frac{1}{2}} \left\{ A + \left(\frac{Bx + C}{1 + x^2} \right) \times (1 + 2x) \right\},$$

$$\frac{\frac{1}{4} + 2 - 1}{1 + \frac{1}{4}} = A,$$

$$A = 1.$$

$$\lim_{x \rightarrow i} \left\{ \frac{x^2 - 4x - 1}{1 + 2x} \right\} = \lim_{x \rightarrow i} \left\{ \left(\frac{A}{1 + 2x} \right) \times (1 + x^2) + Bx + C \right\},$$

$$\frac{-1 - 4i - 1}{1 + 2i} = Bi + C,$$

$$-2 - 4i = Bi + C - 2B + 2iC,$$

$$\therefore C - 2B = -2,$$

$$B + 2C = -4,$$

$$2C - 4B = -4,$$

$$5B = 0,$$

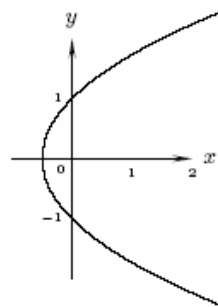
$$B = 0,$$

$$C = -2.$$

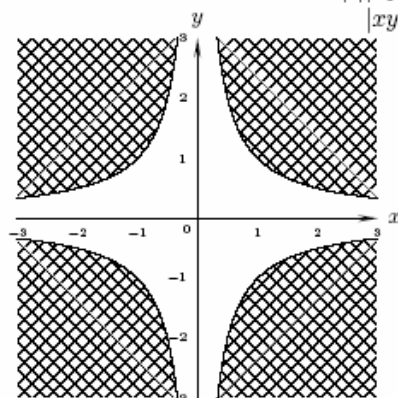
2

$$\begin{aligned} \text{(ii) } I &= \frac{1}{2} \int \frac{2x dx}{1 + 2x} - 2 \int \frac{dx}{1 + x^2}, \\ &= \frac{1}{2} \ln(1 + 2x) - 2 \tan^{-1} x + C. \end{aligned}$$

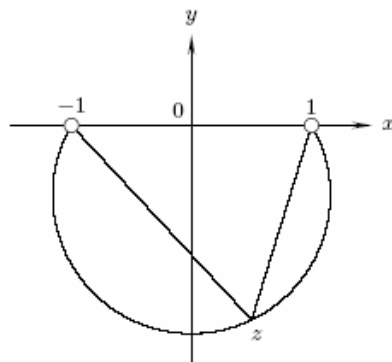
2. (a) (i) $2\sqrt{x^2 + y^2} = x + iy + x - iy + 2,$
 $x^2 + y^2 = x^2 + 2x + 1,$
 $y^2 = 4\left(\frac{1}{2}\right)\left(x + \frac{1}{2}\right).$



(ii) $|(x + iy)^2 - (x - iy)^2| \geq 4,$
 $|x^2 + 2ixy - y^2 - (x^2 - 2ixy - y^2)| \geq 4,$
 $|4xy| \geq 4,$
 $|xy| \geq 1.$



(iii)



$$\boxed{2} \quad \text{(b) (i) } \overrightarrow{BA} = (10 - 6) + i(2 - 8), \\ = 4 - 6i.$$

Note that the question was in error: what was meant was $\overrightarrow{AB}$. Both answers were accepted, $4 - 6i$ or $-4 + 6i$.

$$\boxed{2} \quad \text{(ii) Method 1: } \overrightarrow{AC} = \overrightarrow{AB} \times 2 \times \text{cis } \frac{\pi}{4}, \\ = -(4 - 6i) \times 2 \times \left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} \right), \\ = -(4\sqrt{2} + 4\sqrt{2}i - 6\sqrt{2}i + 6\sqrt{2}), \\ = -10\sqrt{2} + 2\sqrt{2}i. \\ \therefore C = (10 + 2i) + (-10\sqrt{2} + 2\sqrt{2}i), \\ = 10(1 - \sqrt{2}) + 2(1 + \sqrt{2})i.$$

$$\text{Method 2: } \overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AB} \times 2 \times \text{cis } \frac{\pi}{4}, \\ = (10 + 2i) - (4 - 6i) \times 2 \times \left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} \right), \\ = (10 + 2i) - (4\sqrt{2} + 4\sqrt{2}i - 6\sqrt{2}i + 6\sqrt{2}), \\ = 10 + 2i - 10\sqrt{2} + 2\sqrt{2}i. \\ = 10(1 - \sqrt{2}) + 2(1 + \sqrt{2})i.$$

$$\boxed{1} \quad 3. \quad \text{(a) (i) Roots } \alpha, \beta \text{ positive implies } \alpha\beta > 0, \\ \text{i.e. } \frac{c}{a} > 0 \text{ or } K > 0. \\ \text{Also, for distinct real roots, } \Delta = 1 - 4K > 0, \\ 1 > 4K, \\ K < \frac{1}{4}. \\ \text{So, } 0 < K < \frac{1}{4}.$$

$$\boxed{2} \quad \text{(ii) } \alpha + \beta = 1, \\ \alpha\beta = K, \\ \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta, \\ = 1 - 2K.$$

Method 1:
 $K < \frac{1}{4},$
 $\therefore \alpha^2 + \beta^2 > 1 - 2\left(\frac{1}{4}\right)$ ("greater than" as we are subtracting "less than"),
i.e. $\alpha^2 + \beta^2 > \frac{1}{2}.$

Method 2:
 $2K = 1 - (\alpha^2 + \beta^2),$
 $K = \frac{1 - (\alpha^2 + \beta^2)}{2} < \frac{1}{4},$
 $-(\alpha^2 + \beta^2) < -\frac{1}{2},$
 $\therefore \alpha^2 + \beta^2 > \frac{1}{2}.$

$$\boxed{2} \quad \text{(iii)} \quad \frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2 + \beta^2}{\alpha^2 \beta^2},$$

$$= \frac{1 - 2K}{K^2}, \text{ (from above).}$$

Now, also from above,

$$\alpha^2 + \beta^2 > \frac{1}{2},$$

$$\alpha^2 \beta^2 < \left(\frac{1}{4}\right)^2,$$

$$\frac{1}{\alpha^2 \beta^2} > 16.$$

$$\text{So } \frac{1}{\alpha^2} + \frac{1}{\beta^2} > 16 \times \frac{1}{2},$$

$$\therefore \frac{1}{\alpha^2} + \frac{1}{\beta^2} > 8.$$

$$\boxed{3} \quad \text{(b) (i)} \quad \omega = 2 \left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} \right),$$

$$= 2 \text{cis } \frac{\pi}{4},$$

$$\omega^4 = (2 \text{cis } \frac{\pi}{4})^4,$$

$$= 16 \text{cis } \pi,$$

$$= -16 + 16 \times 0,$$

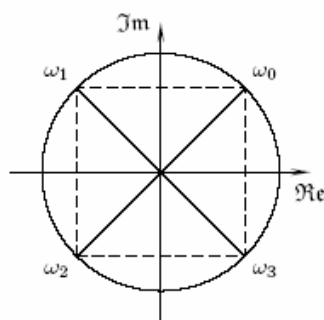
$$= -16.$$

$$\therefore \omega_0 = \sqrt{2} + \sqrt{2}i,$$

$$\omega_1 = -\sqrt{2} + \sqrt{2}i,$$

$$\omega_2 = -\sqrt{2} - \sqrt{2}i,$$

$$\omega_3 = \sqrt{2} - \sqrt{2}i.$$



$$\boxed{2} \quad \text{(ii)} \quad z^4 + 16 = (z - \sqrt{2} - \sqrt{2}i)(z - \sqrt{2} + \sqrt{2}i)(z + \sqrt{2} + \sqrt{2}i)(z + \sqrt{2} - \sqrt{2}i),$$

$$= (z^2 - 2\sqrt{2}z + 4)(z^2 + 2\sqrt{2}z + 4).$$

$$\boxed{2} \quad \text{(iii) Method 1:}$$

$$\omega = 2 \text{cis } \frac{\pi}{4},$$

$$\omega^3 = 2 \text{cis } \frac{3\pi}{4},$$

$$\omega^5 = 32 \text{cis } \frac{5\pi}{4},$$

$$\omega^7 = 128 \text{cis } \frac{7\pi}{4}.$$

$$\omega + \frac{\omega^3}{4} + \frac{\omega^5}{16} + \frac{\omega^7}{64} = 2 \text{cis } \frac{\pi}{4} + 2 \text{cis } \frac{3\pi}{4} + 2 \text{cis } \frac{5\pi}{4} + 2 \text{cis } \frac{7\pi}{4},$$

$$= \omega_0 + \omega_1 + \omega_2 + \omega_3, \text{ the sum of the roots,}$$

$$= 0.$$

Method 2:

$$\frac{64\omega + 16\omega^3 + 4\omega^5 + \omega^7}{64} = (16\omega(4 + \omega^2) + \omega^5(4 + \omega^2)) \times \frac{1}{64},$$

$$= \omega(16 + \omega^4)(4 + \omega^2) \times \frac{1}{64}.$$

But $16 + \omega^4 = 0$,

$$\therefore \omega + \frac{\omega^3}{4} + \frac{\omega^5}{16} + \frac{\omega^7}{64} = 0.$$

Method 3:

$$\omega^4 = -16,$$

$$\begin{aligned}\omega + \frac{\omega^3}{4} + \frac{\omega^5}{16} + \frac{\omega^7}{64} &= \omega + \frac{\omega^3}{4} + \frac{-16\omega}{16} + \frac{-16\omega^3}{64}, \\ &= \omega - \omega + \frac{\omega^3}{4} - \frac{\omega^3}{4}, \\ &= 0.\end{aligned}$$

Method 4:

In the geometric series given, $a = \omega$ and $r = \frac{\omega^2}{4}$.

$$\begin{aligned}S_4 &= \frac{\omega \left(1 - \left(\frac{\omega^2}{4} \right)^4 \right)}{1 - \frac{\omega^2}{4}}, \text{ note that } \frac{\omega^8}{256} = \frac{(-16)^2}{256} = 1, \\ &= \frac{\omega(1-1)}{1 - \frac{\omega^2}{4}}, \\ &= 0.\end{aligned}$$

$$(4)(a)(i) \int_0^a x\sqrt{a-x} \, dx$$

$$= \int_a^0 (a-u)\sqrt{u}(-du)$$

$$= \int_0^a (a-u)u^{1/2} \, du$$

$$= \int_0^a (au^{1/2} - u^{3/2})du$$

$$= \left[\frac{2a}{3}u^{3/2} - \frac{2}{5}u^{5/2} \right]_0^a$$

$$= \left(\frac{2a^2}{3} - \frac{2a^2}{5} \right) \sqrt{a}$$

$$= \frac{4a^2}{15} \sqrt{a} = \frac{4a^{5/2}}{15}$$

$$u = a - x \Rightarrow x = a - u; dx = -du$$

$$x = 0 \Rightarrow u = a$$

$$x = a \Rightarrow u = 0$$

$$(ii) \int_0^1 \frac{\sin^{-1} x}{\sqrt{1+x}} \, dx$$

$$= \int_0^1 \left((1+x)^{-1/2} \times \sin^{-1} x \right) dx$$

$$= 2\sqrt{1+x} \sin^{-1} x \Big|_0^1 - \int_0^1 \frac{2\sqrt{1+x}}{\sqrt{1-x^2}} \, dx$$

$$= \sqrt{2}\pi - 2 \int_0^1 \frac{1}{\sqrt{1-x}} \, dx$$

$$= \sqrt{2}\pi + 2 \int_0^1 -(1-x)^{-1/2} \, dx$$

$$= \sqrt{2}\pi + 2 \times 2\sqrt{1-x} \Big|_0^1$$

$$= \sqrt{2}\pi - 4$$

$$(b)(i) \int_0^1 \frac{(2dt/1+t^2)}{1+(1-t^2/1+t^2)+(2t/1+t^2)}$$

$$= \int_0^1 \frac{2dt}{1+t^2+1-t^2+2t}$$

$$= \int_0^1 \frac{2dt}{2+2t}$$

$$= \int_0^1 \frac{dt}{1+t}$$

$$= \left[\ln|1+t| \right]_0^1$$

$$= \ln 2$$

$$t = \tan \frac{x}{2}$$

$$\cos x = \frac{1-t^2}{1+t^2}, \sin x = \frac{2t}{1+t^2}$$

$$dx = \frac{2dt}{1+t^2}$$

4 (b) (ii)

$$\begin{aligned}
 I &= \int_0^{\frac{\pi}{2}} \frac{x dx}{1 + \sin x + \cos x} \\
 &= \int_{\frac{\pi}{2}}^0 \frac{-\left(\frac{\pi}{2} - u\right) du}{1 + \sin\left(\frac{\pi}{2} - u\right) + \cos\left(\frac{\pi}{2} - u\right)} \\
 &= \int_0^{\frac{\pi}{2}} \frac{\left(\frac{\pi}{2} - u\right) du}{1 + \cos u + \sin u}
 \end{aligned}$$

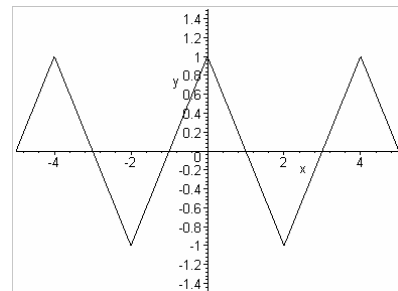
$$\begin{aligned}
 u &= \frac{\pi}{2} - x \Rightarrow x = \frac{\pi}{2} - u; dx = -du \\
 x = 0 &\Rightarrow u = \frac{\pi}{2}; x = \frac{\pi}{2} \Rightarrow u = 0
 \end{aligned}$$

$$\begin{aligned}
 \therefore 2I &= \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2} du}{1 + \cos u + \sin u} = \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \frac{du}{1 + \cos u + \sin u} = \frac{\pi}{2} \times \ln 2 \\
 \therefore I &= \frac{\pi \ln 2}{4}
 \end{aligned}$$

5 (a)

(i) $y = h(x+1)$

Move the curve 1 unit to the left



(ii) $y = \frac{1}{h(x)}$

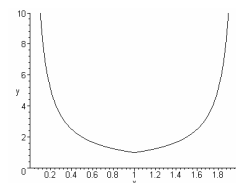
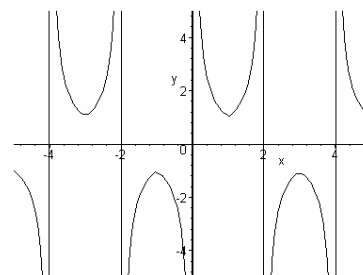
Where $h(x) = 0$ there are vertical asymptotes.

Where $h \rightarrow 0^+$, $y \rightarrow \infty$

Where $h \rightarrow 0^-$, $y \rightarrow -\infty$

Where $h = 1$, the reciprocal is *pointed* ie not smooth.

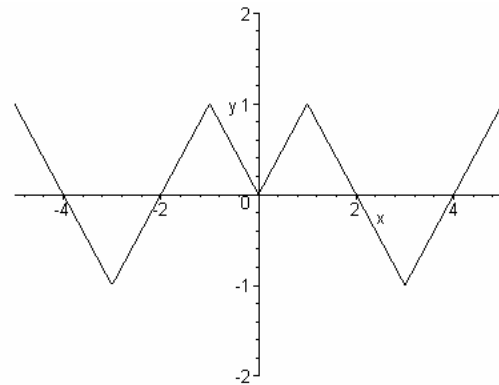
(See the bottom diagram on the right.)



5 (a)

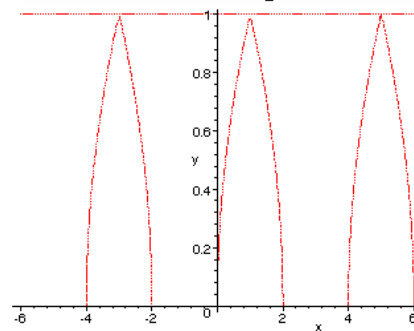
(iii) $y = h(|x|)$.

Erase the LHS of h and then reflect the RHS, so that the result is an even function.



(iv) $y = \sqrt{h(x)}$

First erase the graph where $h < 0$.
Where $0 < h < 1 \Rightarrow \sqrt{h} > h$
Where $y = 1$, the graph is *pointed*, ie not smooth.
Where $y = 0$, vertical tangents.



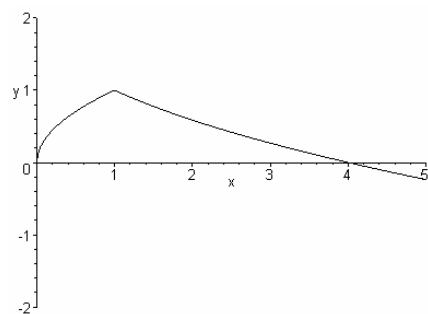
(v) $y = h(\sqrt{x})$

Domain: $x \geq 0$

Note that $0 \leq x \leq 4 \Rightarrow 0 \leq \sqrt{x} \leq 2$

So $h(\sqrt{4}) = h(2) = 0$

The graph for $0 \leq x \leq 4$ will be the same y values for h over $0 \leq x \leq 2$.



5 (b)

First draw $9y = x(x-3)^2$

Clearly x intercepts are at $x = 0$ and $x = 3$ **with** $x = 3$ is a double root.

$$y = x(x-3)^2/9 \Rightarrow z' = \frac{1}{9}(2x(x-3) + (x-3)^2) = \frac{(x-3)}{9}(2x+x-3)$$

$$\therefore y' = \frac{1}{9}(x-3)(3x-3) = \frac{1}{3}(x-1)(x-3) = \frac{1}{3}(x^2 - 4x + 3)$$

$$\therefore y'' = \frac{1}{3}(2x-4)$$

Stationary points when $y' = 0 \Rightarrow x = 1, 3$ ie $(1, \frac{4}{9})$ & $(3, 0)$

At $x = 1$, $y'' < 0 \Rightarrow (1, \frac{4}{9})$ is a maximum.

The graph in Fig I is the graph of z . The horizontal line is the line $y = 1$.

So with $y = \frac{1}{3}\sqrt{x(x-3)^2}$, the maximum turning point remains the same except it is now $(1, \frac{2}{3})$.

Any part of the graph in Fig I below the x - axis is not defined for the square root.

Where $0 < y < 1$ we get $\sqrt{y} > y$ and where $y > 1$ we get $\sqrt{y} < y$.

The $x = 0$ intercept will have a vertical tangent, the $x = 3$ intercept is not smooth. This is shown in Fig 2.

We need to draw $y = \pm \frac{1}{3}\sqrt{x(x-3)^2}$: the $\pm$ means that the top part of the graph will be reflected.

The final graph is Fig 3. With turning points $(1, \pm \frac{2}{3})$

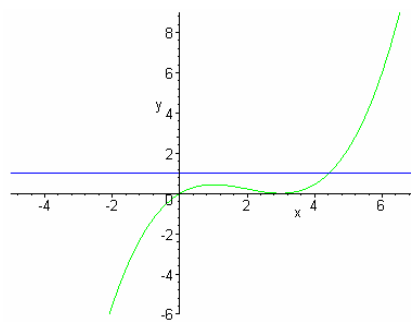


Fig I

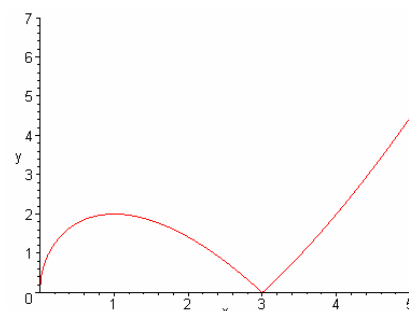


Fig II

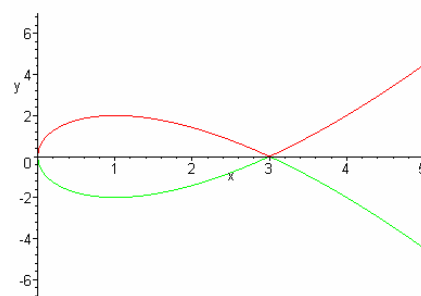
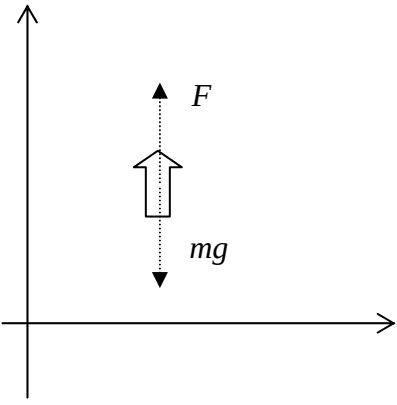
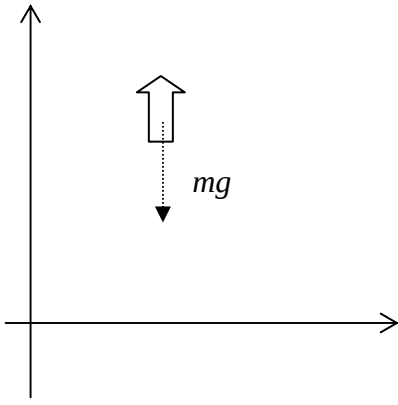


Fig III

Question 6

(i)

$0 \leq t \leq 2$	$t > 2$
	
$F = g(2 - t)$ $\therefore ma = F - mg = g(2 - t) - mg$ $\therefore a = \frac{g(2 - t)}{0.2} - g$ $\therefore a = 5g(2 - t) - g = g(9 - 5t)$	After 2 seconds the only force acting is gravity $\therefore a = -g$

6(ii) (a) For $0 \leq t \leq 2$

$$\frac{dv}{dt} = g(9-5t)$$

$$v = 9gt - \frac{5}{2}gt^2 + C_1$$

$$\left. \begin{matrix} t=0 \\ v=0 \\ C_1=0 \end{matrix} \right\} \therefore v = gt \left[9 - \frac{5}{2}t \right] \quad \text{or } v = \frac{9gt^2}{2} - \frac{5}{6}gt^3 + C_3 \quad (*)$$

Max speed when $a = g(9-5t) = 0$

ie when $t = \frac{9}{5}$

$$\therefore \text{Max speed } v = \frac{81g}{10} \text{ m/s}$$

(B) When $t=2$, $v=8g$

For $t \geq 2$, $a = -g$

$$\frac{dv}{dt} = -g$$

$$v = -gt + C_2$$

$$C_2 = 10g \quad \text{ie } v = 10g - gt \quad (**)$$

For max height $v=0 \Rightarrow \underline{t=10}$

From (*) when $t=2$, $x = \frac{34g}{3}$

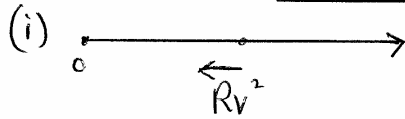
From (**) $x = 10gt - \frac{gt^2}{2} + C_4$

$$\text{When } t=2, x = \frac{34g}{3} \Rightarrow C_4 = -\frac{20g}{3}$$

$$\therefore x = -\frac{gt^2}{2} + 10gt - \frac{20g}{3}$$

$$\text{When } t=10, x = \frac{130g}{3} \text{ m}$$

Question 7



$$m\ddot{x} = -Rv^2$$

$$\text{since } m=1 \Rightarrow \ddot{x} = -Rv^2$$

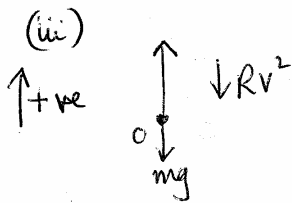
(ii) $\frac{dv}{dt} = -Rv^2$

$$\int_u^v \frac{dv}{v^2} = -R \int_0^t dt$$

$$\left[-\frac{1}{v} \right]_u^v = -Rt \quad 3$$

$$-\frac{1}{v} + \frac{1}{u} = -Rt$$

$$t = \frac{1}{R} \left(\frac{1}{v} - \frac{1}{u} \right)$$



$$m\ddot{y} = -mg - RV^2$$

$$m=1 \Rightarrow \ddot{y} = -(g + RV^2)$$

(iv) $\ddot{y} = -(g + RV^2)$

$$\text{Since } g = Ra^2$$

$$\Rightarrow \ddot{y} = -(Ra^2 + RV^2)$$

$$\ddot{u} \ddot{y} = -R(a^2 + v^2)$$

$$\frac{dw}{dt} = -R(a^2 + v^2)$$

$$\int_u^0 \frac{dw}{a^2 + v^2} = -R \int_0^t dt$$

$$\left[\frac{1}{a} \tan^{-1} \frac{v}{a} \right]_u^0 = -Rt$$

$$0 - \frac{1}{a} \tan^{-1} \frac{u}{a} = -Rt$$

$$\Rightarrow t = \frac{1}{Ra} \tan^{-1} \frac{u}{a} \quad \frac{1}{2}$$

At this time 2nd particle is at rest.

Subst. into (ii) with $v=V$ $\frac{1}{2}$

$$\Rightarrow \frac{1}{R} \left[\frac{1}{V} - \frac{1}{u} \right] = \frac{1}{Ra} \tan^{-1} \frac{u}{a}$$

$$\frac{1}{V} = \frac{1}{u} + \frac{1}{a} \tan^{-1} \frac{u}{a}$$

where V = vel. of 1st particle when the second is momentarily at rest